

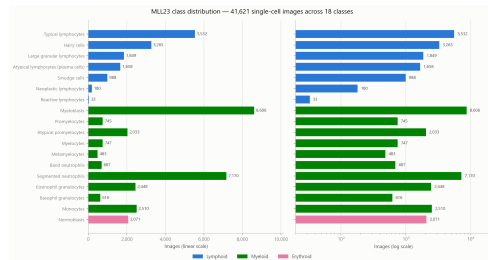
Why this research is needed

The clinical problem

- Reading a blood smear is a core step in diagnosing blood disease, including leukaemia. An expert labels each cell by shape, chromatin, cytoplasm and granularity.
- The manual count is slow, repetitive, and varies between observers — and it is hardest exactly where it matters, on the **rare and immature cells** whose appearance overlaps with their neighbours.

What existing systems get wrong

- They treat all 18 cell types as **flat, unrelated labels**, discarding the known biology: cells belong to lymphoid, myeloid or erythroid lineages, and myeloid types lie on a maturation continuum.
- They report **plain accuracy**, which hides the failure that matters — a model can score 96% while missing most of the rarest class.
- Not all errors are equal: a *within-lineage* confusion is clinically mild, a *cross-lineage* one severe. Flat metrics cannot tell them apart.

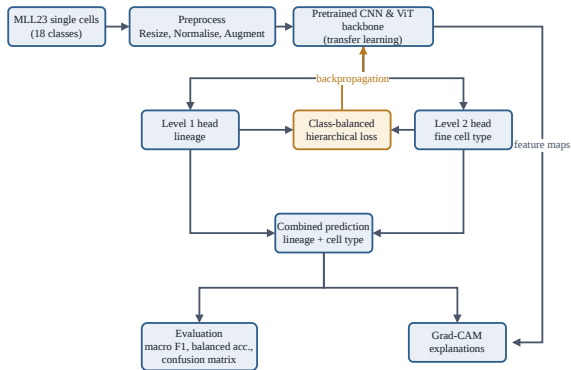


The imbalance is the subject, not an obstacle:
261:1 from myeloblasts (8,606) to reactive lymphocytes (33).

Research question

Does a lineage-aware hierarchical classifier, with an imbalance-robust objective, improve fine-grained classification — **especially for rare cell types** — compared with a flat classifier on the same data?

The proposed method



Two heads, one backbone

Trained jointly, so rare types borrow strength from lineage neighbours. Lineage is a **regulariser**, not a **gate**.

What the arms switch off

One trainer, different flags: no lineage head = **flat baseline**; no class-balanced/focal terms = **imbalance ablation**.

What we have done

Data and analysis **complete**

- 41,621 images acquired, checksum-verified, deterministic 70/15/15 stratified split.
- Decoded-image cache: 178 $\rightarrow$ **~ 850 img/s**. Training was data-bound, not compute-bound.
- Exploratory finding: lineage structure is **local, not global** (silhouette ≈ 0 , but k -NN agreement 0.78 vs 0.45 chance) — so lineage is used as a soft regulariser, not a routing gate.

Method **complete**

- Shared backbone $\rightarrow$ lineage head + fine head, trained jointly under a class-balanced hierarchical loss.
- One config-driven trainer runs every arm, so the ablations cannot drift apart. Grad-CAM and paired significance testing in place.
- Backbone screen selected **ViT**.

Results — v2 matrix, ViT, 30 epochs (8 of 12 runs)

Arm	Seeds	Macro F1	Minority F1	Min. recall
Hierarchical (proposed)	3	0.879	0.840	0.799
Flat baseline	3	0.874	0.823	0.776
No imbalance term	2	0.870	0.771	0.716
Stain-normalised	0	<i>currently training</i>		

- Proposed model wins on every metric, and by more on the rare classes (+0.017) than overall (+0.005) — the predicted pattern. At $n = 3$ seeds, **not yet significant** ($p = 0.29$, $d_z = 0.84$).
- Removing the imbalance term costs **-0.069** minority F1: that objective is the load-bearing component.
- The hierarchy is nearly **inert** — lineage accuracy saturates at 98.9%, so it regularises an axis already solved.
- v1 matrix **invalidated** (run length drove the score, $r = +0.87$); protocol rebuilt, noise **-54%** .

What we do next — 1. Scale up the experiments

Work	Runs	Est. GPU
Finish the v2 matrix (no_imbalance s2, stain_norm $\times 3$)	4	~ 5 h
Raise 3 $\rightarrow$ 5 seeds on all arms	8	~ 10 h
Full backbone comparison at 30 epochs (ResNet / ConvNeXt / MobileNet $\times 3$)	9	~ 12 h
Total	21	~ 27 h

Spent so far: **16.6 GPU-hours** for 8 runs on an RTX 4070 Laptop.

Why this is the priority

Three seeds gives **2 degrees of freedom**. The effect is real in direction and moderate-to-large in size, but the *sample of runs* is too small to establish it. Five seeds, combined with the halved measurement noise, is what turns the paired tests from indicative into informative.

Also outstanding

The 6-epoch screen *selected* a backbone; it is not the backbone *comparison* the proposal promised. Full-length runs are needed to report that honestly.

Decision needed

27 h is feasible locally but strictly serial. A rented cloud GPU would parallelise it into a weekend.

What we do next — 2. Fix the imbalanced-data problem

The problem, stated precisely

Minority recall is stuck at **0.80** — one rare cell in five is still missed. Per-class F1 collapses on reactive lymphocytes (≈ 0.55 , which leaves roughly **23 training images**) and on the adjacent myeloid maturation stages: metamyelocytes, myelocytes, band neutrophils (0.59–0.72). Class-balanced weights plus focal loss is a comparatively weak long-tail lever.

Planned work, cheapest and highest-leverage first:

- 1 **Decoupled training (cRT)**. Learn the representation with instance-balanced sampling, then retrain *only the classifier* with class-balanced sampling. Among the strongest long-tail methods, and it reuses the backbones already trained — minutes per run, not hours.
- 2 **Logit adjustment / LDAM margin loss**. Larger decision margins for rare classes; a drop-in replacement inside the existing `losses.py`.
- 3 **Retarget the auxiliary task** at maturation stage within the myeloid branch, where the residual confusion actually lives, instead of at lineage — which is already solved at 98.9%.

Target

Minority macro F1 0.840 $\rightarrow$ 0.88+ and minority recall off the 0.80 plateau, with $p < 0.05$ at 5 seeds.

Sequence

Finish v2 matrix $\rightarrow$ add cRT and logit adjustment as new arms $\rightarrow$ re-run at 5 seeds $\rightarrow$ results chapter.